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Algorithm 1: Face-Based Finite-Volume Solver Loop for a General Shallow-Water Mesh

Require: Cell states $\mathbf{U}_P^n = [h_P^n, q_{x,P}^n, q_{y,P}^n]^T$, mesh geometry, bathymetry b_P , timestep Δt

Ensure: Updated cell states $\mathbf{U}_P^{n+1}$ and diagnostic outputs

1: Initialise residuals for all cells:

$$\mathbf{R}_P \leftarrow \mathbf{0}$$

2: **for** each cell P **do**

3: Recover primitive variables:

$$u_P = \frac{q_{x,P}}{h_P}, \quad v_P = \frac{q_{y,P}}{h_P}, \quad \eta_P = h_P + b_P$$

4: Apply wet/dry state logic using $h_{\min}$

5: **end for**

6: **for** each face f **do**

7: Identify owner cell P_f and neighbour/ghost state N_f

8: Reconstruct left and right states at the face:

$$\mathbf{U}_{L,f}, \quad \mathbf{U}_{R,f}$$

9: Apply hydrostatic reconstruction to obtain:

$$\mathbf{U}_{L,f}^*, \quad \mathbf{U}_{R,f}^*$$

10: Compute the numerical normal flux:

$$\hat{\mathbf{F}}_{n,f} = \mathcal{F}(\mathbf{U}_{L,f}^*, \mathbf{U}_{R,f}^*, \mathbf{n}_f)$$

11: Apply positivity or draining-time flux limiting where required

12: Accumulate the owner-cell residual:

$$\mathbf{R}_{P_f} \leftarrow \mathbf{R}_{P_f} - A_f \hat{\mathbf{F}}_{n,f}$$

13: **if** f is an internal face **then**

14: Accumulate the neighbour-cell residual:

$$\mathbf{R}_{N_f} \leftarrow \mathbf{R}_{N_f} + A_f \hat{\mathbf{F}}_{n,f}$$

15: **end if**

16: **end for**

17: **for** each cell P **do**

18: Assemble the finite-volume right-hand side:

$$\mathcal{L}(\mathbf{U}_P) = \frac{\mathbf{R}_P}{|\Omega_P|} + \mathbf{S}_P$$

19: Advance the state using the selected time integrator:

$$\mathbf{U}_P^{n+1} = \mathcal{T}(\mathbf{U}_P^n, \mathcal{L}(\mathbf{U}_P), \Delta t)$$

20: Apply post-update wet/dry and positivity corrections

21: Update cell diagnostics and output quantities

22: **end for**

$$\mathcal{C}_P = \begin{cases} \mathcal{G}_P = \{I_P, \mathbf{x}_P, |\Omega_P|, \partial\Omega_P, \mathcal{N}(P), \tau_P, \chi_P\}, & \text{cell geometry} \\ \mathcal{U}_P = \{h_P, q_{x,P}, q_{y,P}\}, & \text{conserved shallow-water state} \\ \mathcal{B}_P = \{b_P, \eta_P, \nabla b_P, b_{\min,P}, b_{\max,P}\}, & \text{bathymetry and free-surface data} \\ \mathcal{M}_P = \{n_P, f_P, \nu_{H,P}, h_{\min,P}, \kappa_P\}, & \text{local source-model parameters} \\ \mathcal{R}_P = \{R_{h,P}, R_{q_x,P}, R_{q_y,P}, \dot{h}_P, \dot{q}_{x,P}, \dot{q}_{y,P}\}, & \text{residual and right-hand-side data} \\ \mathcal{D}_P = \{h_{\max,P}, \eta_{\max,P}, |\mathbf{u}|_{\max,P}, t_{\text{arrival},P}, t_{\text{wet},P}, \text{CFL}_P\}, & \text{diagnostic and visualisation data} \end{cases} \quad (0.1)$$

$$\mathcal{F}_f = \begin{cases} \mathcal{I}_f = \{I_f, P_f, N_f\}, & \text{face indexing and connectivity} \\ \mathcal{G}_f = \{\mathbf{x}_f, A_f, \mathbf{n}_f, \mathbf{t}_f\}, & \text{face geometry} \\ \mathcal{S}_f = \{\mathbf{U}_{L,f}, \mathbf{U}_{R,f}, \mathbf{U}_{L,f}^*, \mathbf{U}_{R,f}^*\}, & \text{left, right, and reconstructed states} \\ \mathcal{Q}_f = \{u_{n,L}, u_{n,R}, c_L, c_R, S_L, S_R\}, & \text{wave-speed and normal-flow data} \\ \mathcal{H}_f = \{b_{L,f}, b_{R,f}, b_f^*, h_{L,f}^*, h_{R,f}^*\}, & \text{hydrostatic reconstruction data} \\ \mathcal{R}_f = \{\hat{\mathbf{F}}_{n,f}, A_f \hat{\mathbf{F}}_{n,f}, \theta_f\}, & \text{normal flux and limited flux contribution} \\ \mathcal{B}_f = \{\beta_f, \gamma_f, \sigma_f\}, & \text{boundary, aperture, and mask data} \end{cases} \quad (0.2)$$

Here, I_P and I_f denote cell and face indices, respectively; $\mathbf{x}_P$ and $\mathbf{x}_f$ are the cell and face centroids; $|\Omega_P|$ is the cell area; A_f is the face length in two dimensions; $\mathbf{n}_f$ is the outward unit normal from owner cell P_f to neighbour cell N_f ; and $\mathbf{t}_f$ is the corresponding tangential vector. The variable τ_P denotes the cell topology, such as triangular, quadrilateral, polygonal, or cut-cell, while χ_P stores the cell-region classification, such as ocean, land, coastline, harbour, or defence region. The set $\mathcal{N}(P)$ stores the neighbouring cells connected to P through its faces.

The conserved state is stored as $\mathcal{U}_P = \{h_P, q_{x,P}, q_{y,P}\}$, where h_P is the water depth and $q_{x,P} = h_P u_P$, $q_{y,P} = h_P v_P$ are the depth-integrated momenta. The bathymetric block $\mathcal{B}_P$ stores the bed elevation b_P , free-surface elevation $\eta_P = h_P + b_P$, bed-gradient data, and local bed extrema used by wetting, drying, and hydrostatic reconstruction. The source-model block $\mathcal{M}_P$ stores Manning roughness n_P , Coriolis parameter f_P , horizontal eddy viscosity $\nu_{H,P}$, wet/dry threshold $h_{\min,P}$, and a roughness or material class κ_P .

The residual block $\mathcal{R}_P$ stores the accumulated finite-volume flux imbalance and the resulting right-hand-side components used by the time integrator. The diagnostic block $\mathcal{D}_P$ stores quantities required for scientific interpretation and numerical-health monitoring, including maximum depth, maximum surface elevation, maximum speed, arrival time, wetting duration, and local CFL number.